

FRACTIONS ALL OPERATIONS, (ADDING AND SUBTRACTING FRACTIONS -

Name: _____

Date: _____

LO: I can add and subtract fractions with different denominators by finding a common denominator.

Adding and subtracting fractions

To add or subtract fractions with different denominators, we must first find a common denominator so that both fractions have the same-sized parts.

Remember: Find the lowest common multiple (LCM) of the denominators, convert both fractions, then add or subtract the numerators.

Quick examples

●	$\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$	LCM of 3 and 4 is 12
●	$\frac{3}{5} - \frac{1}{3} = \frac{4}{15}$	LCM of 5 and 3 is 15
●	$\frac{2}{3} + \frac{1}{6} = \frac{5}{6}$	LCM of 3 and 6 is 6
●	$\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$	cannot add numerators and denominators

Key idea

Find the LCM of the denominators, convert each fraction to equivalent fractions, then add or subtract numerators.

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}$$

Cross-multiply to find equivalent numerators

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - adding unit fractions

Calculate $\frac{1}{3} + \frac{1}{4}$

LCM of 3 and 4 = 12

$$\frac{1}{3} = \frac{4}{12} \text{ and } \frac{1}{4} = \frac{3}{12}$$

$$\frac{4}{12} + \frac{3}{12} = \frac{7}{12}$$

$$= \frac{7}{12}$$

Convert both fractions to twelfths, then add numerators.

Example 2 - subtracting fractions

Calculate $\frac{3}{4} - \frac{2}{5}$

LCM of 4 and 5 = 20

$$\frac{3}{4} = \frac{15}{20} \text{ and } \frac{2}{5} = \frac{8}{20}$$

$$\frac{15}{20} - \frac{8}{20} = \frac{7}{20}$$

$$= \frac{7}{20}$$

Same process: find common denominator, then subtract.

Example 3 - adding with simplifying

Calculate $\frac{2}{3} + \frac{1}{6}$

LCM of 3 and 6 = 6

$$\frac{2}{3} = \frac{4}{6}$$

$$\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$$

$$= \frac{5}{6}$$

When one denominator is a factor of the other, only convert one fraction.

Example 4 - mixed numbers

Calculate $1\frac{1}{3} + 2\frac{1}{4}$

Convert: $\frac{4}{3} + \frac{9}{4}$

$$\text{LCM} = 12: \frac{16}{12} + \frac{27}{12}$$

$$= \frac{43}{12} = 3\frac{7}{12}$$

$$= 3\frac{7}{12}$$

Convert mixed numbers to improper fractions first.

YOUR TURN – PRACTICE (deliberate practice)

1. Adding fractions (unit)

Find a common denominator and add.

- a) $\frac{1}{2} + \frac{1}{3}$
- b) $\frac{1}{4} + \frac{1}{5}$
- c) $\frac{1}{3} + \frac{1}{6}$
- d) $\frac{1}{5} + \frac{1}{10}$

2. Adding and subtracting

Calculate. Simplify where possible.

- a) $\frac{2}{3} + \frac{1}{4}$
- b) $\frac{3}{5} - \frac{1}{4}$
- c) $\frac{5}{6} - \frac{1}{3}$
- d) $\frac{3}{8} + \frac{1}{4}$

3. Harder fractions

These involve larger denominators.

- a) $\frac{5}{6} - \frac{2}{9}$
- b) $\frac{7}{8} - \frac{2}{3}$
- c) $\frac{4}{7} + \frac{2}{5}$
- d) $\frac{3}{4} - \frac{5}{12}$

4. Mixed numbers

Add or subtract the mixed numbers.

- a) $1\frac{1}{2} + 2\frac{1}{3}$
- b) $3\frac{1}{4} - 1\frac{1}{5}$
- c) $2\frac{2}{3} + 1\frac{3}{4}$
- d) $4\frac{1}{2} - 2\frac{2}{5}$

5. Mixed practice

Add or subtract. Simplify your answer.

- a) $\frac{2}{5} + \frac{3}{10}$
- b) $\frac{7}{9} - \frac{1}{3}$
- c) $1\frac{3}{8} + 2\frac{1}{6}$
- d) $\frac{5}{6} + \frac{2}{9} - \frac{1}{3}$
- e) $3\frac{1}{3} - 1\frac{3}{4}$

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{3} + \frac{1}{4} = \frac{7}{12}$ ☐

Correct answer: _____

Reason: _____

b) $\frac{1}{3} + \frac{1}{4} = \frac{2}{7}$ ☐

Correct answer: _____

Reason: _____

c) To add fractions we need a common denominator ☐

Correct answer: _____

Reason: _____

d) $\frac{3}{5} - \frac{1}{3} = \frac{2}{2} = 1$ ☐

Correct answer: _____

Reason: _____

e) $\frac{5}{6} - \frac{1}{3} = \frac{5}{6} - \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A recipe needs $\frac{2}{3}$ cup of flour and $\frac{3}{4}$ cup of sugar. How much more sugar than flour is needed? What is the total amount of flour and sugar combined?

Expression: _____

Simplified: _____

b) Aisha walks $1\frac{1}{4}$ km to school and $1\frac{2}{3}$ km to her friend's house. How far does she walk in total? How much further is it to her friend's house than to school?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

